

PADHAISHURU

GATE Preparation Platform

GATE EE

Electrical Engineering

TREND-BASED MOCK PAPER

— 04 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	11 subjects

Difficulty Distribution

Easy: 11 questions (17%)

Moderate: 44 questions (68%)

Difficult: 10 questions (15%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE EE — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[General Aptitude]

Choose synonym of 'pragmatic':

- (A) Idealistic
- (B) Practical
- (C) Theoretical
- (D) Abstract

Q2. 1M [-0.33]

[General Aptitude]

Choose opposite of 'latent':

- (A) Obvious
- (B) Hidden
- (C) Clear
- (D) Manifest

Q3. 1M [no -ve]

[General Aptitude]

Distance 200km at 50km/h takes:

Enter your answer here

Q4. 1M [-0.33]

[General Aptitude]

Speed = Distance /:

- (A) Time
- (B) Velocity
- (C) Acceleration
- (D) Force

Q5. 1M [-0.33]

[Engineering Mathematics]

Eigenvalues of identity:

- (A) All zero
- (B) All one
- (C) Distinct
- (D) Depends

Q6. 1M [-0.33]

[Electrical Machines]

Slip in induction motor:

- (A) Always positive
- (B) Can be negative
- (C) Always zero
- (D) Infinity

Q7.

1M [-0.33]

[Electrical Machines]

DC starter used to:

- (A) Regulate speed
- (B) Limit start current
- (C) Reverse
- (D) Increase torque

Q8.

1M [no -ve]

*[Electrical Machines]*Slip $s =$

*Enter your answer here***Q9.**

1M [-0.33]

[Electrical Machines]

Alternator produces:

- (A) DC
- (B) AC
- (C) Both
- (D) None

Q10.

1M [-0.33]

[Electrical Machines]

Transformer works on:

- (A) Auto induction
- (B) Mutual induction
- (C) Self induction
- (D) None

Q11.

1M [-0.33]

[Power Systems]

Synchronous generator excitation:

- (A) Rotor
- (B) Stator
- (C) Both
- (D) None

Q12.

1M [-0.33]

[Power Systems]

Most efficient transmission:

- (A) Low voltage
- (B) High voltage
- (C) DC
- (D) AC

Q13.

1M [-0.33]

[Power Systems]

Circuit breaker operates on:

- (A) Overload
- (B) Short circuit
- (C) Both
- (D) None

Q14.

1M [-0.33]

[Power Systems]

Per unit system base values:

- (A) Voltage only
- (B) Power only
- (C) Voltage and Power
- (D) Current

Q15.

1M [-0.33]

[Control Systems]

Gain margin:

- (A) At phase crossover
- (B) At gain crossover
- (C) Both
- (D) None

Q16.

1M [-0.33]

[Control Systems]

Root locus plots:

- (A) Open loop poles
- (B) Closed loop poles
- (C) Both
- (D) None

Q17.

1M [-0.33]

[Control Systems]

Integral controller eliminates:

- (A) Rise time
- (B) Steady state error
- (C) Damping
- (D) Overshoot

Q18.

1M [-0.33]

[Control Systems]

Time constant of RC:

- (A) RC
- (B) R/C
- (C) C/R
- (D) 1/RC

Q19.

1M [-0.33]

[Power Electronics]

Step-down chopper output:

- (A) Less than input
- (B) More than input
- (C) Equal
- (D) Zero

Q20.

1M [-0.33]

[Power Electronics]

Inverter converts:

- (A) DC to AC
- (B) AC to DC
- (C) DC to DC
- (D) AC to AC

Q21.

1M [-0.33]

[Power Electronics]

Thyristor is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q22.

1M [-0.33]

[Power Electronics]

Full-wave rectifier uses:

- (A) 1
- (B) 2
- (C) 4
- (D) 3

Q23.

1M [no -ve]

[Analog Electronics]

Ideal op-amp gain:

*Enter your answer here***Q24.**

1M [-0.33]

[Analog Electronics]

RC phase shift uses:

- (A) 180°
- (B) 360°
- (C) 90°
- (D) 45°

Q25.

1M [-0.33]

[Analog Electronics]

Negative feedback:

- (A) Increases gain
- (B) Decreases gain
- (C) No effect
- (D) Oscillates

Q26.

1M [-0.33]

[Analog Electronics]

ICBO in transistor is:

- (A) Collector-Base reverse
- (B) Emitter-Base
- (C) Collector-Emitter
- (D) None

Q27.

1M [no -ve]

[Digital Electronics]

PLD stands for:

*Enter your answer here***Q28.**

1M [-0.33]

[Digital Electronics]

Universal gate:

- (A) AND
- (B) OR
- (C) NAND
- (D) XOR

Q29.

1M [-0.33]

[Digital Electronics]

Mod-10 counts from:

- (A) 0 to 10
- (B) 0 to 9
- (C) 1 to 10
- (D) 1 to 9

Q30.

1M [no -ve]

[Signals and Systems]

Aliasing avoided by:

*Enter your answer here***Q31.**

2M [-0.66]

[Signals and Systems]

Nyquist rate for 4kHz signal:

- (A) 4kHz
- (B) 8kHz
- (C) 2kHz
- (D) 1kHz

Q32.

2M [-0.66]

[Signals and Systems]

FS coefficients:

- (A) Discrete
- (B) Continuous
- (C) Both
- (D) None

Q33.

2M [-0.66]

[EMFT]

Displacement current added by:

- (A) Faraday
- (B) Maxwell
- (C) Ampere
- (D) Gauss

Q34.

2M [no -ve]

[EMFT]

Corona loss increases with:

*Enter your answer here***Q35.**

2M [-0.66]

[Electric Circuits]

Power factor =

- (A) R/Z
- (B) X/Z
- (C) Z/R
- (D) Z/X

Q36.

2M [-0.66]

[Power Electronics]

Full-wave rectifier uses:

- (A) 1
- (B) 2
- (C) 4
- (D) 3

Q37.

2M [-0.66]

[Control Systems]

Root locus plots:

- (A) Open loop poles
- (B) Closed loop poles
- (C) Both
- (D) None

Q38.

2M [no -ve]

[Signals and Systems]

LTI characterized by:

*Enter your answer here***Q39.**

2M [no -ve]

[Signals and Systems]

Aliasing avoided by:

*Enter your answer here***Q40.**

2M [-0.66]

[Electrical Machines]

Alternator produces:

- (A) DC
- (B) AC
- (C) Both
- (D) None

Q41.

2M [-0.66]

[Digital Electronics]

JK flip-flop toggles when:

- (A) $J=K=0$
- (B) $J=K=1$
- (C) $J=0, K=1$
- (D) $J=1, K=0$

Q42.

2M [-0.66]

[Electric Circuits]

Impedance of RL series:

- (A) $R + j\omega L$
- (B) $R - j\omega L$
- (C) $R + 1/j\omega L$
- (D) $R - 1/j\omega L$

Q43.

2M [no -ve]

[Signals and Systems]

Aliasing avoided by:

*Enter your answer here***Q44.**

2M [-0.66]

*[Electrical Machines]*Slip $s =$

- (A) $(N_s - N_r)/N_s$
- (B) $(N_r - N_s)/N_s$
- (C) N_s/N_r
- (D) N_r/N_s

Q45.

2M [-0.66]

[Power Systems]

Synchronous generator excitation:

- (A) Rotor
- (B) Stator
- (C) Both
- (D) None

Q46.

2M [no -ve]

[Electrical Machines]

DC starter used to:

*Enter your answer here***Q47.**

2M [-0.66]

[Power Electronics]

Inverter converts:

- (A) DC to AC
- (B) AC to DC
- (C) DC to DC
- (D) AC to AC

Q48.

2M [no -ve]

[Digital Electronics]

JK flip-flop toggles when:

*Enter your answer here***Q49.**

2M [no -ve]

[Control Systems]

Time constant of RC:

*Enter your answer here***Q50.**

2M [-0.66]

[Power Systems]

Most efficient transmission:

- (A) Low voltage
- (B) High voltage
- (C) DC
- (D) AC

Q51.

2M [no -ve]

[Electric Circuits]

Norton equivalent to:

Enter your answer here

Q52.

2M [no -ve]

[Digital Electronics]

Universal gate:

*Enter your answer here***Q53.**

2M [-0.66]

[Electrical Machines]

DC starter used to:

- (A) Regulate speed
- (B) Limit start current
- (C) Reverse
- (D) Increase torque

Q54.

2M [-0.66]

[Digital Electronics]

JK flip-flop toggles when:

- (A) $J=K=0$
- (B) $J=K=1$
- (C) $J=0, K=1$
- (D) $J=1, K=0$

Q55.

2M [-0.66]

[General Aptitude]

Speed = Distance /:

- (A) Time
- (B) Velocity
- (C) Acceleration
- (D) Force

Q56.

2M [-0.66]

[Power Electronics]

Inverter converts:

- (A) DC to AC
- (B) AC to DC
- (C) DC to DC
- (D) AC to AC

Q57.

2M [no -ve]

[Power Electronics]

Full-wave rectifier uses:

*Enter your answer here***Q58.**

2M [-0.66]

[Analog Electronics]

Negative feedback:

- (A) Increases gain
- (B) Decreases gain
- (C) No effect
- (D) Oscillates

Q59.

2M [no -ve]

[Electric Circuits]

Norton equivalent to:

- (A) Thevenin
- (B) Superposition
- (C) Max power
- (D) KCL

Q60.

2M [no -ve]

[Power Systems]

Circuit breaker operates on:

- (A) Overload
- (B) Short circuit
- (C) Both
- (D) None

Q61.

2M [-0.66]

[Electric Circuits]

Impedance of RL series:

- (A) $R + j\omega L$
- (B) $R - j\omega L$
- (C) $R + 1/j\omega L$
- (D) $R - 1/j\omega L$

Q62.

2M [-0.66]

[Power Electronics]

Thyristor is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q63.

2M [-0.66]

[Power Systems]

Synchronous generator excitation:

- (A) Rotor
- (B) Stator
- (C) Both
- (D) None

Q64.

2M [-0.66]

[Digital Electronics]

PLD stands for:

- (A) Programmable Logic Device
- (B) Processor Logic
- (C) Programmable Logic Design
- (D) None

Q65.

2M [-0.66]

[Signals and Systems]

LTI characterized by:

- (A) Impulse response
 - (B) Step response
 - (C) Frequency
 - (D) All
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